Scrambled states survive local Pauli noise up to the hashing entropy

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Scrambling spreads quantum information over high-weight Pauli operators, which local noise damps exponentially, so scrambled encodings are expected to be fragile. We show that they survive up to a sharp threshold. Two pure states drawn independently from any state 2-design, such as random stabilizer states, stay distinguishable after one layer of i.i.d. Pauli noise until the error entropy reaches one bit per qubit, then become indistinguishable through a Gaussian crossover. For depolarizing noise this hashing threshold, $r = 0.252$, lies far below the $r = 0.423$ reachable by purity bounds.

Introduction.—Random quantum states model the output of scrambling dynamics and of generic quantum circuits. A state 2-design, an ensemble whose first two moments match those of Haar-random states, captures their low-order statistics; random stabilizer states are a standard example [1]. Such states are delocalized: in the Pauli basis almost all of their weight sits on operators acting on a constant fraction of the qubits. Local depolarizing noise of strength r multiplies a weight-w Pauli operator by $(1-r)^w$, so a scrambled state loses almost all of its Pauli weight to noise. This picture underlies the exponential cost of quantum error mitigation (QEM): Takagi, Tajima, and Gu [2] and Tsubouchi, Sagawa, and Yoshioka [3] proved that recovering ideal expectation values needs a number of samples exponential in circuit depth [4, 5], and Quek *et al.* [6] showed that random circuits reach this regime at very shallow depth.

Here we ask how much information a scrambled encoding retains after one layer of noise, and find a sharp answer. Two pure states ψ_1, ψ_2 are drawn independently from a state 2-design and both pass through the same Pauli channel $\mathcal{N}$. An observer who knows the two candidates measures the output to decide which one was prepared. With equal priors the optimal success probability is $(1+\overline{T})/2$, where

$$\overline{T} = \mathbb{E}_{\psi_1,\psi_2} \tfrac{1}{2} \|\mathcal{N}(\psi_1) - \mathcal{N}(\psi_2)\|_1. \tag{1}$$

We prove that, at every system size, $\overline{T}$ is bracketed by a single classical quantity: the probability that the Pauli error has surprisal below n. For i.i.d. noise this places a sharp threshold at the hashing entropy, one bit per qubit, with a universal Gaussian crossover. For local depolarizing noise the threshold is $r_c = 0.252$: up to this strength the encoded states remain almost perfectly distinguishable, even though the noise has already damped their Pauli weight exponentially.

Two observations put this in context. First, the noise must not be followed by another random unitary. Averaging $U^\dagger \mathcal{N}(U \cdot U^\dagger) U$ over a unitary 2-design turns any Pauli channel into global depolarization $\rho \mapsto \theta\rho + (1-\theta) I/d$ with $\theta = (d^2 p_I - 1)/(d^2-1)$ [7], which decays exponentially in n for every $r > 0$. Without this twirl, the optimal measurement uses the local structure of the noise. At $n = 8$ and $r = 0.2$ the distance is 0.736, against about 0.272 after twirling [Fig. 1(a)]. Second, purity cannot see the threshold. The purity argument behind Ref. [6] treats all errors alike and certifies loss only for $(1-r)^2 < 1/3$, i.e., $r > 0.423$. Ibarrondo and Stilck França [8] and Liu *et al.* [9] obtained the same two values, 0.25 from below and 0.42 from above, and left the interval between them open. Our bound closes it. Sommers *et al.* [10] found the same hashing transition, with an $n^{-1/2}$ window, in the coherent information of random codes.

Information-spectrum bracket.—Let $d = 2^n$, $\mathcal{P}_n = \{I, X, Y, Z\}^{\otimes n}$, and

$$\mathcal{N}_p(\rho) = \sum_{P \in \mathcal{P}_n} p_P P \rho P, \qquad s(P) = -\log_2 p_P, \tag{2}$$

with p an arbitrary distribution, so errors may be correlated, and with the surprisal s distributed as $P \sim p$. Logarithms are base two. Each $\psi_i = |\psi_i\rangle\langle\psi_i|$ is drawn independently from an exact state 2-design on the full space,

$$\mathbb{E}\psi_i = I/d, \qquad \mathbb{E}\psi_i^{\otimes 2} = (I+F)/[d(d+1)], \tag{3}$$

where F swaps the two copies; the two ensembles may differ.

Theorem 1. *For every Pauli distribution p and every $a \geq 0$,*

$$\mathbb{P}[s < n-a] - 2^{-a} \leq \overline{T} \leq \mathbb{P}[s < n+a] + 2^{-a/2}/\sqrt{2}. \tag{4}$$

More generally, for any set $\mathcal{G} \subseteq \mathcal{P}_n$ and any partition $\{\mathcal{C}_j\}$ of $\mathcal{P}_n$, with $m_j = \sum_{\mathcal{C}_j} p_P$ and $v_j = \sum_{\mathcal{C}_j} p_P^2$,

$$\overline{T} \geq p(\mathcal{G}) - |\mathcal{G}|/d, \tag{5}$$

$$\overline{T} \leq \sum_j \min\left\{ m_j, \sqrt{\frac{d^2 v_j - m_j^2}{2(d+1)}} \right\}. \tag{6}$$

Proof. Fix ψ_1 and let Π project onto the span of $\{P|\psi_1\rangle : P \in \mathcal{G}\}$. For each realization, $\frac{1}{2}\|\mathcal{N}_p(\psi_1) - \mathcal{N}_p(\psi_2)\|_1 \geq \operatorname{tr}\Pi[\mathcal{N}_p(\psi_1) - \mathcal{N}_p(\psi_2)]$. The first term is at least $p(\mathcal{G})$. Averaged over the independent ψ_2, the second is $\operatorname{tr}\Pi/d \leq |\mathcal{G}|/d$, using only $\mathbb{E}\psi_2 = I/d$. Averaging over ψ_1 gives (5). This is the typical-image argument of Ref. [8], applied to an arbitrary set of Pauli errors.

For the upper bound let $\Delta = \psi_1 - \psi_2$ and $\mathcal{N}_j(\rho) = \sum_{\mathcal{C}_j} p_P P\rho P$. Pauli orthogonality and (3) give

$$\mathbb{E}\,\mathrm{tr}\,\mathcal{N}_j(\psi_i)^2 = \frac{dv_j + m_j^2}{d+1}, \qquad \mathbb{E}\,\mathrm{tr}\,\mathcal{N}_j(\psi_1)\mathcal{N}_j(\psi_2) = \frac{m_j^2}{d},$$
$$\mathbb{E}\,\mathrm{tr}\,\mathcal{N}_j(\Delta)^2 = \frac{2d}{d+1}v_j - \frac{2m_j^2}{d(d+1)}. \tag{7}$$

Jensen's inequality and $\|A\|_1 \le \sqrt{d}\|A\|_2$ bound $\mathbb{E}\frac{1}{2}\|\mathcal{N}_j(\Delta)\|_1$ by the square root in (6), and $\|\Delta\|_1 \le 2$ bounds it by m_j; summing over j gives (6). For (4), take $\mathcal{G} = \{s < n - a\}$, which has fewer than $d2^{-a}$ elements, and split the partition at $s = n + a$, where the tail has $v \le 2^{-n-a}$. □

The theorem has a simple reading. The error images $P|\psi_1\rangle$ of a scrambled state are nearly orthogonal, and a d-dimensional space holds at most d of them. An error with $p_P > 1/d$, i.e., $s(P) < n$, is likely enough to be accommodated, and the projector onto these images separates ψ_1 from an independent ψ_2. The remaining errors are too numerous to fit, and their contributions average out. The distinguishability therefore equals the probability of a likely error, up to a window of width a in surprisal and exponentially small corrections.

Taking the trivial partition, $\mathcal{C} = \mathcal{P}_n$, reduces the upper bound (6) to the purity of the noisy output, $\overline{T} \le [d\sum_P p_P^2/2]^{1/2}$, the quantity that controls the QEM bounds of Ref. [6]. For local depolarizing noise $\sum_P p_P^2 = [(1 + 3(1-r)^2)/4]^n$, which beats d^{-1} only for $r > 1 - 1/\sqrt{3} = 0.423$. The purity bound is loose below that value because it spreads the likely errors over the whole space. Separating them at surprisal n removes this looseness and lowers the threshold to the hashing point.

Sharp threshold.—For i.i.d. noise, $p = \boldsymbol{p}^{\otimes n}$ with a single-qubit distribution $\boldsymbol{p} = (p_0, p_x, p_y, p_z)$. The surprisal is then a sum of n independent terms with mean nH, where $H = \mathbb{E}[-\log_2 p_i]$ is the entropy of $\boldsymbol{p}$, and variance $n\sigma^2$, where $\sigma^2 = \mathrm{Var}[-\log_2 p_i]$. Taking $a = \epsilon n$ in (4) with a large-deviation bound [11] gives

$$\overline{T} = \begin{cases} 1 - e^{-\Omega(n)}, & H < 1, \\ e^{-\Omega(n)}, & H > 1. \end{cases} \tag{8}$$

The threshold $H = 1$ is the hashing point, where the coherent information $1 - H$ of the Pauli channel at maximally mixed input vanishes [12, 13]. For depolarizing noise $\mathcal{N}_r(\rho) = (1-r)\rho + rI/2$, with $q = 3r/4$ and $\boldsymbol{p} = (1-q, q/3, q/3, q/3)$, it lies at $q_c = 0.189290\ldots$, or $r_c = 0.252386\ldots$. (Degenerate codes can transmit quantum information slightly beyond the hashing point [14]; r_c is the threshold for distinguishing two encoded states.)

The bracket also resolves the transition. Let the single-qubit distribution $\boldsymbol{p}_n$ depend on n, with i.i.d. errors at each n, and define $x_n = \sqrt{n}(1 - H_n)/\sigma_n$.

Corollary 1. *If $\boldsymbol{p}_n \to \boldsymbol{p}_*$ with $H(\boldsymbol{p}_*) = 1$ and $\sigma(\boldsymbol{p}_*) > 0$, then for every pair of independent exact 2-design ensembles,*

$$\overline{T}_n = \Phi(x_n) + O\left(\frac{\log n}{\sqrt{n}}\right), \tag{9}$$

where Φ is the standard normal distribution function.

Proof. The Berry–Esseen theorem [15] gives $\mathbb{P}[s < n \mp a] = \Phi(x_n \mp a/\sqrt{n}\sigma_n) + O(n^{-1/2})$, uniformly because the third moment of the surprisal is bounded on a four-letter alphabet and σ_n is bounded away from zero. Insert into (4) with $a = 2\log_2 n$ and use the bounded density of Φ. □

Near threshold the distinguishability thus depends on r only through $\sqrt{n}(r - r_c)$ and follows the universal profile Φ. Figure 1 shows the finite-size bracket at $n = 8$, exact Haar averages, and the collapse onto Φ at large n.

Above threshold the distance decays exponentially. Grouping errors into empirical types in (6) gives $\overline{T}_n \le \mathrm{poly}(n)2^{-nE(\boldsymbol{p})}$, with [11]

$$E(\boldsymbol{p}) = \min_t \{D(t\|\boldsymbol{p}) + \tfrac{1}{2}[H(t) - 1]_+\}$$
$$= \max_{0\le s\le 1} \frac{s}{1+s}[H_{1+s}(\boldsymbol{p}) - 1], \tag{10}$$

where D is the relative entropy and H_α the Rényi entropy. This equals the universal$_2$ privacy-amplification exponent [16] and follows without the polynomial prefactor from decoupling without smoothing [17], since the family $X \mapsto \sum_P X_{PP} P\psi P$ is 1-randomizing under (3). Decoupling [9, 17] therefore also yields decay for $H > 1$; the bracket supplies the location of the transition and its Gaussian profile. Near threshold the lower bound (5), applied to types with $D(t\|\boldsymbol{p}) + H(t) < 1$, matches E to leading order. Writing $R_\pm$ for the upper and lower limits of $-n^{-1}\log_2 \overline{T}_n$, both satisfy

$$R_\pm = \kappa(r - r_c)^2 + O((r - r_c)^3),$$
$$\kappa = \frac{9}{32\ln 2\, q_c(1 - q_c)} = 2.64\ldots \tag{11}$$

Correlated noise.—The bracket applies to any Pauli distribution, and for correlated errors the full information spectrum, rather than the entropy, determines the answer. Global depolarization $\mathcal{N}(\rho) = \theta\rho + (1-\theta)I/d$ is a Pauli channel with $p_I = \theta + (1-\theta)/d^2$ and $p_P = (1-\theta)/d^2$ otherwise. Its entropy rate approaches $2(1-\theta)$, above one for $\theta < 1/2$, yet $\mathcal{N}(\psi_1) - \mathcal{N}(\psi_2) = \theta(\psi_1 - \psi_2)$ gives $\overline{T} \to \theta$. The bracket captures this exactly: with $a = n/2 - 1$ only the identity lies below surprisal $n - a$, all other errors lie above $n + a$, and $|\overline{T} - \theta| \le 2^{-n/4+1}$.

Design dependence.—The bracket depends on the ensemble only through its first two moments, so the threshold and the Gaussian profile are the same for every 2-design. Two finer features depend on higher moments.

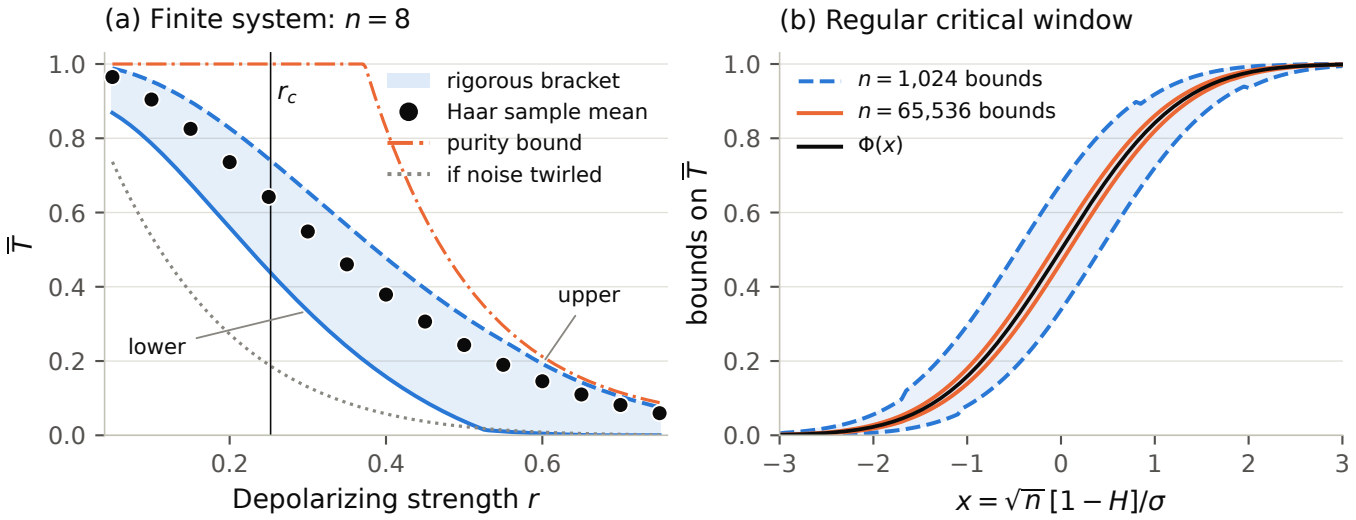


FIG. 1. Distinguishability after one layer of local depolarizing noise. (a) At $n = 8$, the shaded band lies between the projector lower bound (solid) and the two-class upper bound (dashed) of Eqs. (5) and (6), optimized over error-weight cutoffs; the thin vertical line marks r_c. Points are averages over 64 independent Haar-state pairs from full density matrices, with one-standard-error bars; the same pairs are used at every r. The dash-dotted curve is the purity bound, which drops below one only for $r \gtrsim 0.37$ at this size and certifies decay only above 0.423 asymptotically. The dotted gray curve is the depolarizing parameter θ reached if a further 2-design unitary twirled the noise; it equals the twirled distance up to a factor $1 - O(1/d)$. The lower bound is zero for $r \geq 2/3$, where no single error has probability above $1/d$. (b) The bracket (4) with $a = 2\log_2 n$, evaluated from binomial distributions at fixed $x = \sqrt{n}(1 - H)/\sigma$ for $n = 1024$ and 65536, approaches $\Phi(x)$ (black).

The first is the limit at a degenerate critical point. Complete dephasing, $\boldsymbol{p}_* = (1/2, 0, 0, 1/2)$, has $H = 1$ but $\sigma = 0$, so Corollary 1 does not apply. The uniform union of a complete set of $d+1$ mutually unbiased bases (MUBs) is an exact 2-design [18]. Taking the computational basis as one of them, its d states survive dephasing and all others map to I/d, so

$$\overline{T}_{\mathrm{MUB}} = \frac{(d-1)(2d+1)}{d(d+1)^2} \to 0, \qquad \overline{T}_{\mathrm{Haar}} = \frac{d-1}{2d-1} \to \frac{1}{2}. \tag{12}$$

For Haar states the dephased outputs are diagonal; their trace distance is the total-variation distance of two independent flat Dirichlet vectors, and $\mathbb{E}\min(X, Y) = (2d-1)^{-1}$ for single coordinates $X, Y \sim \mathrm{Beta}(1, d-1)$. Two ensembles with identical second moments thus reach different limits (Fig. 2), refining the observation of Ref. [8] that dephasing leaves computational-basis states uncontracted. The variance condition must also hold in the limit. With $\epsilon_n = n^{-6}$ and $\boldsymbol{p}_n = (1/2 - \epsilon_n + \delta_n, \epsilon_n, \epsilon_n, 1/2 - \epsilon_n - \delta_n)$, where $\delta_n = O(\sqrt{\epsilon_n \log(1/\epsilon_n)})$ solves $H(\boldsymbol{p}_n) = 1$, the n-qubit channel differs from complete dephasing by at most $n\|\boldsymbol{p}_n - \boldsymbol{p}_*\|_1 \to 0$ in diamond norm. MUB inputs still give $\overline{T}_n \to 0$, although $\sigma_n > 0$ for every n.

The second is the decay rate away from threshold. A random stabilizer state is $\psi = d^{-1}[I + \sum_{P \neq I} z_P P]$ with $z_P \in \{0, \pm 1\}$ and $\mathbb{E}|z_P| = 1/(d+1)$. Local depolarization multiplies the weight-w terms by $(1 - r)^w$, so

$$\overline{T}_n^{\mathrm{stab}} \leq \frac{[1 + 3(1 - r)]^n - 1}{2^n + 1}. \tag{13}$$

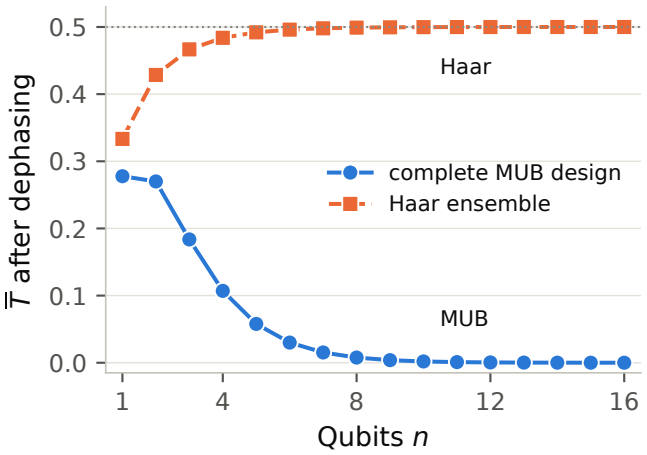


FIG. 2. Design dependence at a degenerate critical point. Under complete dephasing both ensembles are exact 2-designs and the error entropy is one bit per qubit, yet Eq. (12) gives different limits. The dotted gray line marks 1/2.

At $r = 0.9$ this decays at rate at least $1 - \log_2(1.3) = 0.621$, faster than the rate $E = 0.479$ of Eq. (10). Random stabilizer states form a 3-design [1], so the decay rate depends on the ensemble beyond three moments. The Haar rate away from threshold remains open.

Discussion.—Scrambling does not make encoded information fragile to local Pauli noise. Two scrambled states remain distinguishable up to the hashing entropy, where the error distribution carries one bit per qubit, and the

transition has a universal Gaussian shape. One bracket, built from the error distribution and the Hilbert-space dimension alone, controls this for every 2-design and for correlated noise.

The result sharpens the information-theoretic view of noisy quantum computation. QEM costs are exponential at large depth for any noise strength [2, 3, 6], and the purity estimates behind those bounds treat all errors alike. After a single noisy layer, the optimal measurement exploits the likely errors, which these estimates miss; this moves the threshold from 0.423 to 0.252. Splitting the error distribution at surprisal n is a direct way to sharpen purity-based bounds in other settings. Natural next steps are worst-case versions for QEM, deeper circuits with noise between layers, approximate designs from shallow circuits, and non-Pauli noise.

Acknowledgments.—Generative AI was used substantively in this work. Claude Opus 5.5 (Anthropic), accessed through Claude Code in October 2026, with one review pass by another Claude model, was directed by the author to select the problem, draft and revise the derivations, survey the literature, write the analysis code, generate the figures, and draft the text. A review with an OpenAI GPT model [version to be specified by the author] identified the omitted Ref. [8]. The author is responsible for all content. Proofs were rederived independently, the exact counterexamples were checked by explicit enumeration for $n \leq 2$, numerical claims were checked against independent implementations recorded with the data, and cited statements were checked against the primary sources.

Data availability.—The data that support the findings of this article and the code that generates every figure and numerical check accompany this submission and will be deposited in a public repository [19].

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